https://www.qbitlabs.co.in/

bit labs
...Go Beyond Realms

TSN ETHERNET SWITCH (QSXG24)

The QBIT TSN Ethernet Switch (QSXG20-MIL) is a managed Time-Sensitive Networking (TSN) switch designed for deterministic Ethernet networks. It provides 4 x 10GBASE-SFP ports and 16 x 10/100/1000BASE-T(X) TSN ports along with a console interface, enabling high-bandwidth aggregation and endpoint connectivity in converged industrial and mission-critical networks.

The switch integrates key IEEE 802.1 TSN capabilities for precise time synchronization, scheduled traffic, per-stream policing and frame replication/elimination for reliability. Comprehensive Layer-2 and Layer-3 features, along with secure configuration and lifecycle management, simplify deployment in safety and mission-critical environments.

Introduction

Time-Sensitive Networking (TSN) extends standard Ethernet with deterministic latency, bounded jitter and precise time synchronization. The TSN Ethernet Switch forms the backbone of TSN domains, providing deterministic forwarding between TSN endpoints while maintaining standard Ethernet interoperability for best-effort traffic.

It supports IEEE 802.1AS gPTP timing, IEEE 802.1Qbv time-aware scheduling, IEEE 802.1Qci per-stream filtering and policing and IEEE 802.1CB frame replication and elimination for reliability (FRER). Integrated Layer-3 routing, multicast control, redundancy mechanisms and secure management interfaces enable deployment in converged networks with mixed TSN and non-TSN traffic.

Architecture

This TSN Ethernet switch delivers deterministic, low-latency communication with 16 × 1G ports and 4 × 10G ports. All 20 ports connect through high-speed SerDes lanes to a central Assembler/Deassembler block. The architecture splits into two main parts: the TSN data-plane subsystem and an integrated ARM/RISC-V CPU subsystem. The TSN engine features ingress classification, PSFP policing, FRER redundancy, IEEE 802.1AS timestamping, and advanced QoS. A unified shared memory pool and non-blocking switching fabric handle traffic across all ports efficiently.

Deterministic behavior is ensured via time-aware shaping (802.1Qbv), frame preemption, and automated scheduled frame injection. On egress, frames undergo modification, filtering, and precise scheduling before transmission. The CPU subsystem includes timers, sensors, eMMC, DDR memory, Frame DMA, and standard interfaces (UART, SPI, GPIO, JTAG). Overall, the design combines high bandwidth, full TSN feature set, and flexible physical-layer options for converged real-time networks.

Why Choose Us?

🚀 Rapid integration, faster time-to-market

📘 Clean, verified, documented RTL

🤝 Direct access to experts

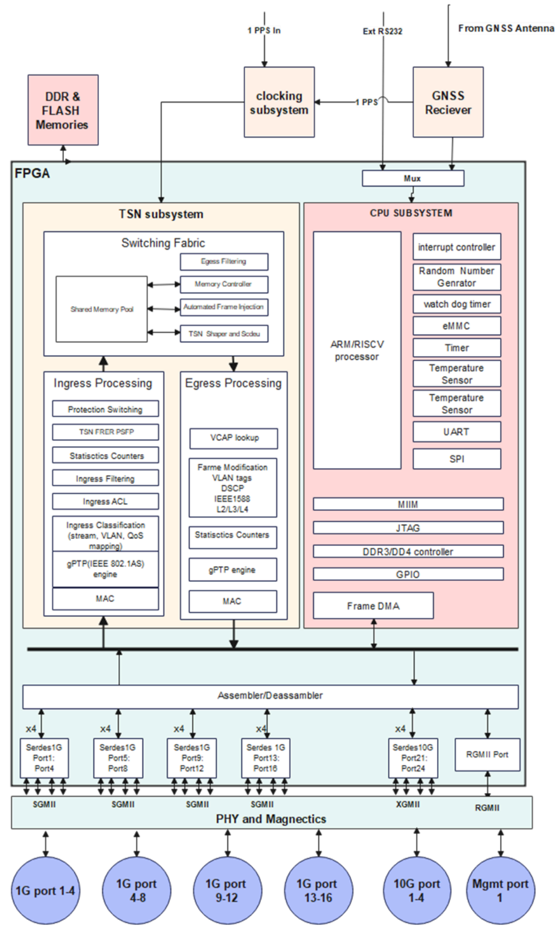
104, FF, Iris Tech Park, Sector 48
Sohna Road, Gurugram, 122018

+91 124 4048731

sales@qbitlabs.co.in

Features

- 4 x 10GBASE-SFP TSN Ethernet ports.
- 16 x 10/100/1000BASE-T(X) TSN Ethernet ports. The ports can be configured to be TSN/non-TSN compliant.
- 1 PPS input, RS232 and 28V DC power port.
- Switching capacity is 100 Gbps with wire-speed forwarding for 64-9216 byte frames; jumbo frame support.
- IEEE 802.1AS timing and synchronization (gPTP).
- IEEE 802.1Qbv Time Aware Shaper (TAS) with scheduled traffic; status monitoring of TSN streams.
- IEEE 802.1Qci per-stream filtering and policing; IEEE 802.1CB Frame Replication and Elimination for Reliability (FRER).
- IEEE 802.3-2008 compliant Ethernet switching with automatic MAC address learning/aging and static MAC table.
- **VLAN support:** port-based VLANs, priority tagged/untagged frames and IEEE 802.1Q VLAN tagging (up to 4K VLAN groups) with shared or independent VLAN learning.
- **QoS support:** IEEE 802.1p CoS/QoS and Layer-3/4 QoS classification (e.g., DSCP) for best-effort IP traffic.
- **Multicast support:** IGMP snooping (v1/v2/v3) for IPv4, MLD snooping for IPv6, Layer-2 multicast filtering and support for at least 255 multicast groups.
- Secure management: HTTPS web interface, SSHv2 CLI, SNMPv3, Syslog eventing, DHCP support, and encrypted & digitally signed firmware/bitstream upgrades with save/restore configuration.



Specification

Parameters	Specifications
Interface Configuration	• 4 x 10GBASE-R TSN Ethernet ports • 16 x 10/100/1000BASE-T(X) TSN Ethernet ports • 1x 50 ohms PPS output • **Miscellaneous connector Port** (i) 1x RS-422 PPS output (ii) 1x 50 ohms PPS input (iii) 1x RS-422 PPS input (iv) 2x RS-232 serial line (v) 1x 10MHz input • Power supply 28VDC
Performance Specification	• Switching capacity: minimum 100 Gbps • Wire-speed performance across all packet sizes from 64-9216 bytes • Jumbo frame support
TSN features	• IEEE 802.1AS - Timing and synchronization (gPTP) • IEEE 802.1Qbv - Time Aware Shaper (TAS) • IEEE 802.1Qci - Per-stream filtering and policing • IEEE 802.1CB - Frame Replication and Elimination for Reliability (FRER) • Status monitoring of TSN stream

Specification Cont...

Layer 2 General Functionalities	• IEEE 802.3-2008 (Ethernet) • Automatic MAC address learning and aging; static MAC table • Port-based VLANs; IEEE 802.1Q VLAN tagging (up to 4K VLAN groups) • IEEE 802.1p CoS/QoS; IEEE 802.1AB (LLDP) • Storm control for flooded broadcast, multicast and unicast • Spanning Tree Protocol: IEEE 802.1D (STP), IEEE 802.1w (RSTP), IEEE 802.1s (MSTP) • Tagged, priority tagged and untagged frame handling with configurable VLAN tag insertion • Shared and independent VLAN learning
Synchronization	• Time reference **1.** obtained from an internally populated GNSS chipset (external antenna required through the antenna connector) supporting GPS L1, Galileo, GLONASS G1 and Bei Dou B1 frequencies **2.** External 1 PPS and NEMA message over UART • Battery backup high accuracy RTC (+ 3 ppm/250ms per day) • gPTP synchronization accuracy to the reference clock **1.** + 40 ns (typical) **2.** +25ns (for GNSS reference) • NTP: Server/Client. **1.** NTP v4 (RFC 5905) and backward compatible with lower versions **2.** SNTP (REC 4330).
Multicast	• Support for at least 255 multicast groups • IGMP snooping (v1, v2 and v3) for IPv4 • Layer 2 multicast filtering • IGMP snooping (up to 255 multicast filters)
Security Features	• IEEE 802.1X for port based network access control. • IEEE 802.1AE MACsec (Media Access Control Security) support. • Manual MAC addresses filtering mechanism. • Selective ports disabling capability. • Unsecure protocols disabling capability. • MAC port binding & authentication for login security. • Secure Shell (SSH) Protocol v2 for command line interface. • TPM 2.0 IC for identity authentication. • Encryption/authentication & signature for firmware and bitstream.
Configuration & Management	• HTTPS web interface • SSHv2 command line interface (CLI) • Encrypted and digitally signed firmware/bitstream upgrades • Saving and restoring configuration • Internal status monitoring and logging • Event notification through Syslog • Statistics independent per port • SNMPv3 • DHCP support

104, FF, Iris Tech Park, Sector 48
Sohna Road, Gurugram, 122018
+91 124 4048731
sales@qbitlabs.co.in

Specification Cont...

High Availability	• Link Aggregation protocol IEEE 802.3ad (LACP) • MTBF not less than 100,000 hours
Operating Temperature	• Operating Temperature: -40°C to +85°C? • Storage Temperature: -55°C to +105°C • Humidity: 10% to 90% non-condensing
Physical	• Dimensions (mm): 250(W) | 170(D) | 110(H) • Weight: 2.3kg • Copper, fibre optic, power supply and Miscellaneous port via D38999 connector
Electrical	• Power Input: 28VDC • Power consumption: 30W
Compliance	• MIL-STD-461G CE101, CE102, CS101, CS114, CS115, CS116, RE101, RE102, RS101, RS103 • MIL-STD-810G Methods: 500.5, 501.5, 502.5, 506.5, 507.5, 508.6, 509.5, 510.5, 513.6, 514.6, 516.6 • MIL-STD 704

Deliverables

1. Switch HW
2. Mating Cables
3. Power Adapter (100V-240V AC to 28V DC)

Get in touch with us

Delivering Advanced Solutions for Next-Gen Networking

104, FF, Iris Tech Park, Sector 48
Sohna Road, Gurugram, 122018

+91 124 4048731

sales@qbitlabs.co.in